



**Department of Electrical and Computer Engineering
North South University**

Senior Design Project

**SMART HAND TREMOR SUPPRESSION AND
REHABILITATION GLOVE FOR PARKINSON'S
PATIENTS**

SUDIPTO SARKER SUPTO

ID# 2131748043

SHAH MD. ABIR

ID# 2131867043

MD ASHRAFUL ISLAM

ID# 2132700643

Faculty Advisor:

Dr. Fariah Mahzabeen

Assistant Professor

ECE Department

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Approval

Sudipto Sarker Supto (ID# 2131748043), Shah Md. Abir (ID# 2131867043) and Md Ashraful Islam (ID# 2132700643) from Electrical and Computer Engineering Department of North South University, have worked on the Senior Design Project titled “Smart Hand Tremor Suppression and Rehabilitation Glove for Parkinson’s Patients” under the supervision of Dr. Fariah Mahzabeen partial fulfillment of the requirement for the degree of Bachelors of Science in Engineering and has been accepted as satisfactory.

Supervisor’s Signature

.....

Dr. Fariah Mahzabeen

Assistant Professor

Department of Electrical and Computer Engineering

North South University

Dhaka, Bangladesh.

Chairman’s Signature

.....

Dr. Abdul Matin

Professor

Department of Electrical and Computer Engineering

North South University

Dhaka, Bangladesh.

Declaration

This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

Students' names and signatures

1. Sudipto Sarker Supto

.....

2. Shah Md. Abir

.....

3. Md Ashraful Islam

.....

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Abstract

Parkinson's disease (PD) is a progressive neurodegenerative disorder that impairs motor function, with hand tremors being one of the most disruptive symptoms. These tremors interfere with essential daily tasks like eating, writing, and dressing, greatly diminishing patients' independence and quality of life. Current treatment options like medication and therapy provide limited relief, do not offer real-time tremor suppression, and are often expensive.

This project presents the design and development of a Smart Hand Tremor Suppression and Rehabilitation Glove—a wearable, non-invasive solution that addresses both immediate tremor suppression and long-term motor rehabilitation. The glove uses a combination of servo motors, IMU sensors, and flex sensors to detect and counteract tremor activity in real-time. A tension-anchor mechanism powered by servo motors applies restorative force to the fingers via elastic bands when tremors are detected, reducing involuntary motion.

Beyond suppression, the glove includes a rehabilitation mode that guides the user through controlled hand exercises to support motor recovery. What sets this project apart is the integration of machine learning algorithms for pattern recognition and adaptive control. These algorithms are trained to distinguish voluntary from involuntary movements, optimize servo responses, and adapt rehabilitation routines based on user-specific patterns over time, creating a more personalized and intelligent system. The glove also features IoT-based remote monitoring, allowing real-time tracking of tremor severity and rehabilitation progress.

This report details the system design, component integration, experimental results, and analysis of the first working prototype of this project. Initial tests have shown effective tremor detection and suppression response, laying the groundwork for future improvements toward a fully adaptive and user-centered assistive device.

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Chapter 1

Introduction

1.1 Background and Motivation

Parkinson’s disease is a chronic and progressive neurodegenerative disorder that affects millions of people around the world, especially the elderly. One of its most common and troubling symptoms is involuntary hand tremors. These tremors can make simple, everyday tasks—like eating, writing, or buttoning a shirt—feel frustrating or even impossible. Over time, this not only affects a person’s independence but also takes a heavy emotional toll, reducing their overall quality of life.

Although there are medical treatments available, such as Levodopa and deep brain stimulation (DBS), they often have limitations. Some treatments can lose effectiveness over time, and others might not be suitable for all patients due to their side effects or the need for invasive procedures. On top of that, traditional methods of tremor assessment are often subjective, making it harder for doctors to track real-time progress or deterioration. All of this points to a clear need for a better, more personalized approach to managing tremors.

1.2 Purpose and Goal of the Project

This project focuses on designing and developing a Smart Hand Tremor Suppression and Rehabilitation Glove—a wearable solution that aims to go beyond what current treatments offer. The goal is to create a glove that not only helps reduce tremors in real-time using servo-based electromechanical damping but also supports motor rehabilitation by guiding the user through targeted exercises.

The glove uses a combination of servo motors for active tremor suppression, IMU sensors for detecting rapid movements, and flex sensors to monitor finger positioning. By combining these elements, we aim to create a device that reacts instantly to tremor events and adapts to the user’s needs. A unique feature of the glove is its IoT-based monitoring system, which allows caregivers and doctors to track progress remotely and in real time.

To further enhance adaptability and personalization, the system incorporates machine learning algorithms that learn from sensor data over time. These algorithms help differentiate between voluntary and involuntary movements, fine-tune suppression responses, and customize rehabilitation routines to suit the user's condition and progression. This makes the glove smarter with continued use and better suited to long-term therapy.

At its core, the glove is designed to offer more than just symptom management—it aims to promote gradual motor recovery and long-term improvement through active rehabilitation. This dual-purpose design sets our project apart from most existing wearable solutions, which typically focus on either suppression or rehabilitation, but not both.

1.3 Organization of the Report

This report is organized into seven chapters, systematically presenting the design, development, and initial testing of the first prototype of the Smart Hand Tremor Suppression and Rehabilitation Glove.

- **Chapter 1** introduces the project by providing background information, motivation, objectives, and an overview of the goals for the first phase of development.
- **Chapter 2** presents a literature review of existing research related to tremor suppression and rehabilitation technologies, identifying key limitations in current solutions and establishing the motivation for designing a dual-purpose wearable device.
- **Chapter 3** describes the methodology employed during the EEE499A phase, detailing the system design, hardware and software components, and the implementation of the tremor sensing and suppression units for the initial prototype.
- **Chapter 4** discusses the experiments conducted to validate the prototype's functionality, presents the results from simulated tremor testing, and provides an analysis and discussion of the system's performance at this preliminary stage.
- **Chapter 5** evaluates the potential societal, health, safety, legal, and environmental impacts of the project, considering the broader implications even at the prototype level.
- **Chapter 6** outlines the project planning and budgeting process, summarizing the phases completed, resource allocation, and expenditure incurred during the EEE499A phase.

- **Chapter 7** concludes the report by summarizing the achievements of the first prototype, discussing the current limitations, and proposing future improvements and next steps for further development during the EEE499B phase.

Chapter 2

Research Literature Review

Individuals with Parkinson's disease (PD) universally face significant challenges due to hand tremors and motor dysfunctions that severely impair their ability to perform daily tasks, negatively affecting their quality of life [1] - [20]. These tremors, often accompanied by muscle rigidity, weakness, and bradykinesia, are characteristic of PD and persist throughout the disease's progression [15], [20]. Although conventional treatments such as pharmacological therapy and deep brain stimulation (DBS) offer symptom relief, they are often limited by side effects, inconsistent efficacy, and inaccessibility for certain patients, with approximately 25% of individuals remaining unresponsive to standard interventions [1], [2], [8], [11]. Furthermore, traditional tremor assessment methods rely on subjective self-reporting, which lacks the precision and comprehensiveness necessary for effective disease management [20]. Existing therapeutic devices primarily target larger joints such as the wrist and elbow, neglecting the impact of finger tremors on fine motor tasks such as writing and grasping [1], [2], [8], [16]. This gap illustrates the clear need for new approaches that are non-intrusive and fully proactive when it comes to allowing tremor suppression and gradual improvement of functional capabilities [13], [14], [18].

Recent research has increasingly focused on the development of wearable technologies, particularly glove-based solutions, designed to mitigate tremors through mechanical and electromechanical interventions [1]. Several studies have proposed cable-driven systems with actuation mechanisms that counteract tremor-induced movements, achieving amplitude reductions of up to 85% in simulated environments [1], [2] and 73.1%–85.5% across multiple joints in preliminary human trials [8]. Other approaches employ layer-jamming technology, wherein vacuum-induced stiffness in soft materials suppresses tremors by as much as 78.32% while maintaining a lightweight form factor of approximately 30g [3], [6]. Additionally, magnetorheological (MR) fluid-based dampers provide real-time adjustable damping, reducing tremor acceleration and angular velocity by roughly 60% in wrist-worn exoskeletons weighing between 255g and 262g, though their power consumption remains underexplored [5], [7]. Passive vibration absorbers, tuned to tremor frequencies, offer an alternative suppression method, decreasing wrist angular movement by 57.25% without requiring external power

sources [16], [17]. Soft robotic pneumatic actuators with active feedback control demonstrate tremor amplitude reductions of approximately 75%, leveraging accelerometer-based real-time motion tracking, but their bulky designs may compromise user compliance over extended use [4]. Despite these advancements, a major limitation of existing research is the lack of large-scale clinical validation and real-world testing, as many studies rely on simulations or single-patient trials, which hinders generalizability and clinical adoption [1], [2], [4], [8]. While these wearable solutions demonstrate promising tremor suppression capabilities, further investigation is required to determine their effectiveness across varying tremor severities and patient profiles [3], [5]–[7].

Beyond immediate tremor suppression, recent research highlights the potential of wearable devices to support long-term motor rehabilitation [12]. For example, the Smart Glove Rehabilitation Device (SGRD) incorporates soft pneumatic actuators controlled via IoT and Arduino systems to facilitate guided finger movements, aiming to restore fine motor function through repetitive motion training [12]. Similarly, a soft rehabilitation glove equipped with flexible actuators and bending sensors offers personalized training programs with maximum flexion and extension angles, maintaining a safety margin within $\pm 6.5^\circ$ [19]. Gyroscopic-based tremor suppression systems have also demonstrated efficacy, with one study reporting a 92.6% reduction in axial tremors through dynamic modeling and feedback linearization [9]. Moreover, rehabilitation gloves originally designed for post-stroke therapy, such as those utilizing flex sensors to promote movement repetition and intensity, may have applications for PD-related tremor management by facilitating neuroplasticity and motor recovery [10]. However, a critical gap in the literature is the absence of longitudinal studies evaluating the sustained impact of these rehabilitation-focused gloves on PD-specific tremor progression. Many existing rehabilitation devices were initially developed for broader neurological conditions, necessitating further research to optimize their efficacy for PD patients [9], [10], [12], [19]. While these approaches show potential in complementing suppression-based interventions, further exploration is needed to refine training protocols and assess their long-term benefits in PD management [12], [19].

Another crucial aspect of emerging smart glove technology is the integration of real-time tremor monitoring and data collection, which enables both patients and clinicians to track symptom progression and adjust interventions accordingly [20]. Advanced sensor-based gloves utilizing accelerometers and flexible sensors have been shown to quantify tremor severity with 95.83% accuracy, aligning closely with clinical assessments (Kappa values 0.833–0.937) [15].

Systematic reviews of wearable sensors indicate that these devices can accurately detect tremor frequency (e.g., 5.05 ± 2.03 Hz) and motor fluctuations, supporting continuous remote monitoring via IoT platforms [13], [14], [18]. Furthermore, the Smart Glove for PD monitoring integrates accelerometers with smartphone connectivity, enabling neurologists to analyze daily tremor and rigidity metrics, thereby reducing the necessity for frequent in-person consultations [20]. IMU-based monitoring systems, such as those incorporated into tremor suppression orthoses, have demonstrated a 94.4% reduction in power consumption while simultaneously assessing voluntary motion, providing insights into tremor dynamics [11]. Despite these advancements, standardization of monitoring protocols remains a challenge. Many existing systems lack user-friendly interfaces that allow patients to customize settings in real-time, while concerns regarding data privacy and processing reliability persist [11], [14], [18], [20].

Building upon these developments, the proposed Smart Hand Tremor Suppression and Rehabilitation Glove aims to integrate electromechanical damping, assisted rehabilitation, and real-time monitoring into a single wearable device. The tremor suppression mechanism will utilize servo-based electromechanical damping, inspired by existing cable-driven exoskeletons [1], [2], [8] and MR fluid-based dampers [5], [7], targeting suppression rates comparable to the 85%–94.4% reductions reported in previous studies [1], [2], [11]. The rehabilitation module will incorporate flex sensors and guided motion exercises, drawing insights from soft pneumatic rehabilitation devices [9], [12], [19] to promote adaptive motor recovery and neuroplasticity. Real-time tremor monitoring will be facilitated through IMU sensors, offering 95.83% accuracy in tremor detection, similar to accelerometer-based tracking systems [4], [11], [15], [20]. The glove will also feature IoT-based connectivity, allowing for continuous remote monitoring and clinician-assisted adjustments, following the approach used in wearable PD monitoring systems [13], [14], [18], [20]. Furthermore, machine learning algorithms will be incorporated to analyze tremor patterns and optimize rehabilitation strategies, improving the adaptability and effectiveness of the device over time.

Chapter 3

Methodology

3.1 System Design

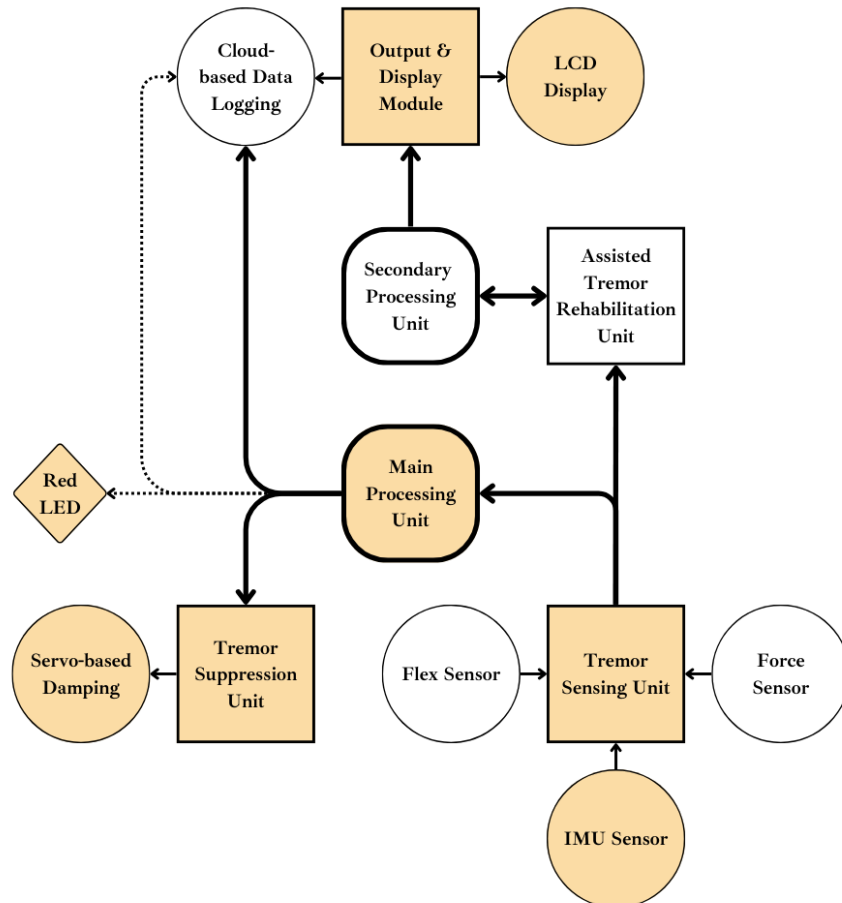


Figure 1: Project block diagram (highlighted boxes are completed during EEE499A)

As shown in Figure 1, the Smart Hand Tremor Suppression and Rehabilitation Glove will consist of six integrated units working together for tremor detection, suppression, and rehabilitation. The Tremor Sensing Unit will be responsible for detecting hand tremors using a combination of IMU, flex, and force sensors. The IMU acts as the primary sensor for detecting tremors, while the flex and force sensors provide additional input to improve accuracy and responsiveness. Together, these sensors offer real-time tracking of hand motion and tremor

intensity. In response to detected tremors, the Tremor Suppression Unit will use servo motors and a tension-anchor mechanism to apply controlled resistance against involuntary hand movements, helping to stabilize the hand actively.

For long-term improvement, the Assisted Tremor Rehabilitation Unit will utilize flex and force sensors to help guide patients through rehabilitation exercises. The Main Processing Unit, built around an Arduino Nano R3, will process tremor data and control suppression. The Secondary Processing Unit, powered by a Raspberry Pi, will manage rehabilitation tracking and cloud-based processing.

Finally, the Output and Display Module will provide real-time feedback to users through an LCD display and IoT-based monitoring system. This module will enable both patients and caregivers to monitor tremor activity, suppression effectiveness, and rehabilitation progress remotely.

The highlighted sections in the block diagram in Figure 1 represent the elements we've that we have successfully implemented during the EEE499A phase of the project.

3.2 Hardware and Software Components

In our Smart Tremor Suppression Glove, we integrated several essential components to bring the system to life. At the core of the project is the Arduino Nano R3, which collects real-time motion data from the MPU6050 inertial measurement unit (IMU) sensor. This sensor tracks the hand's acceleration and rotational movement. To counteract detected tremors, an MG995 servo motor applies controlled tension to the fingers through elastic bands, providing a mechanical damping effect when necessary.

System feedback, such as tremor intensity and operational status, is displayed on an LCD screen for easy monitoring.

Before building the physical prototype, we first designed and simulated the entire circuit virtually using Wokwi, an online simulation platform, to verify component compatibility and ensure correct wiring.

TABLE I: A SAMPLE HARDWARE AND SOFTWARE TOOLS TABLE

Tool	Functions	Other similar Tools (if any)	Why we selected this tool
Arduino Nano	Reads sensor data, processes it, and controls actuators	Arduino Uno, ESP32	Compact, lightweight, and power-efficient — ideal for wearable applications like a glove
MPU6050 IMU Sensor	Detects hand tremors by measuring acceleration and gyroscopic movement	ADXL345	Low cost, reliable performance and provides both accelerometer and gyroscope data in one module
MG995 Servo Motor	Provides mechanical force for tremor suppression by counteracting unwanted hand movements	MG996R	Affordable, high-torque, easily available, and suitable for wrist-mounted actuation
LM2529 Buck Converter	Supplies stable current to servos; protects Arduino Nano and other components from voltage drops	LM7805	Ensures stable operation without damaging sensitive electronics
Mini Vibration motor (MT35)	Simulates tremor effects during testing and validation of the glove	Shaftless Vibration motor(D42)	Compact and easy to integrate into the glove for testing
Battery (Lipo 1500mAh)	Provides portable power supply for the glove (Arduino Nano cannot directly supply enough current to run the MG995)	9V Battery, Power Bank, Li-Ion Battery (18650 Cells)	Lightweight, rechargeable, and capable of handling the higher current demands of servos
Wokwi Simulation Platform	For simulating our Arduino circuits and codes before real-world implementation	Tinker CAD, Proteus	Very easy to use, supports Arduino Nano and MPU6050, allows fast iterations without hardware risk
Arduino IDE	Code development, compilation, and uploading into Arduino Nano	Atmel Studio, Arduino Web Editor	Simple, widely supported, beginner-friendly, with a large library base

On the software side, we programmed the Arduino Nano using the Arduino IDE to continuously read and process motion data from the MPU6050 sensor. A lightweight algorithm was developed to differentiate between rapid, repetitive tremor-like movements typical of Parkinson’s disease and slower, intentional hand motions. When a tremor is detected, the Arduino triggers the servo motor to rotate its horns, tightening the elastic bands and pulling the fingers inward to stabilize the hand. Throughout operation, the LCD display provides real-time updates on tremor detection and glove response, creating a smart feedback loop that allows the glove to react quickly and naturally to the user’s needs.

3.3 Hardware and Software Implementation

Hardware Implementation:

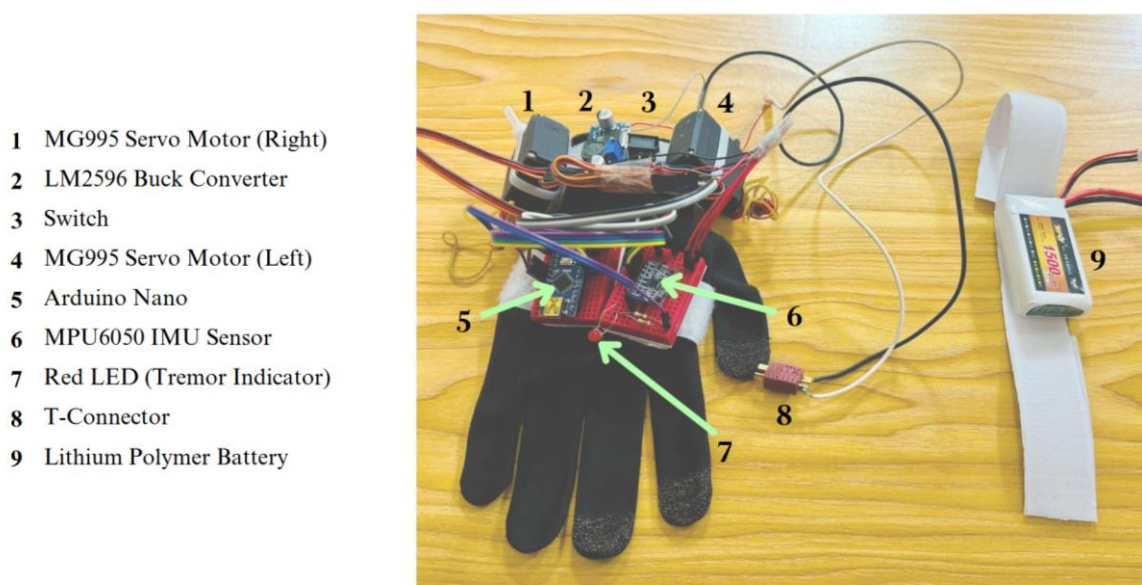


Figure 2: Hardware implementation of our first prototype (Tremor Monitoring and Suppression Units)

The first prototype of the Smart Hand Tremor Suppression and Rehabilitation Glove focused on implementing the Tremor Monitoring and Suppression Units. As illustrated in Figure 3, the hardware employs a tension-anchor mechanism to actively counteract hand tremors. Two MG995 servo motors, mounted on opposite sides of the wrist, pull elastic bands attached to the fingers to generate a controlled damping force. Tremor detection is handled by an MPU6050

IMU sensor positioned on the back of the hand. When rapid involuntary movements are detected, the system responds as follows:

- The left servo horn rotates clockwise by 135° , pulling the thumb and index fingers.
- Simultaneously, the right servo horn rotates counterclockwise by 135° , pulling the pinky and ring fingers.
- The elastic bands, acting like springs, stretch under the servo motion, creating tension that anchors the fingers inward.

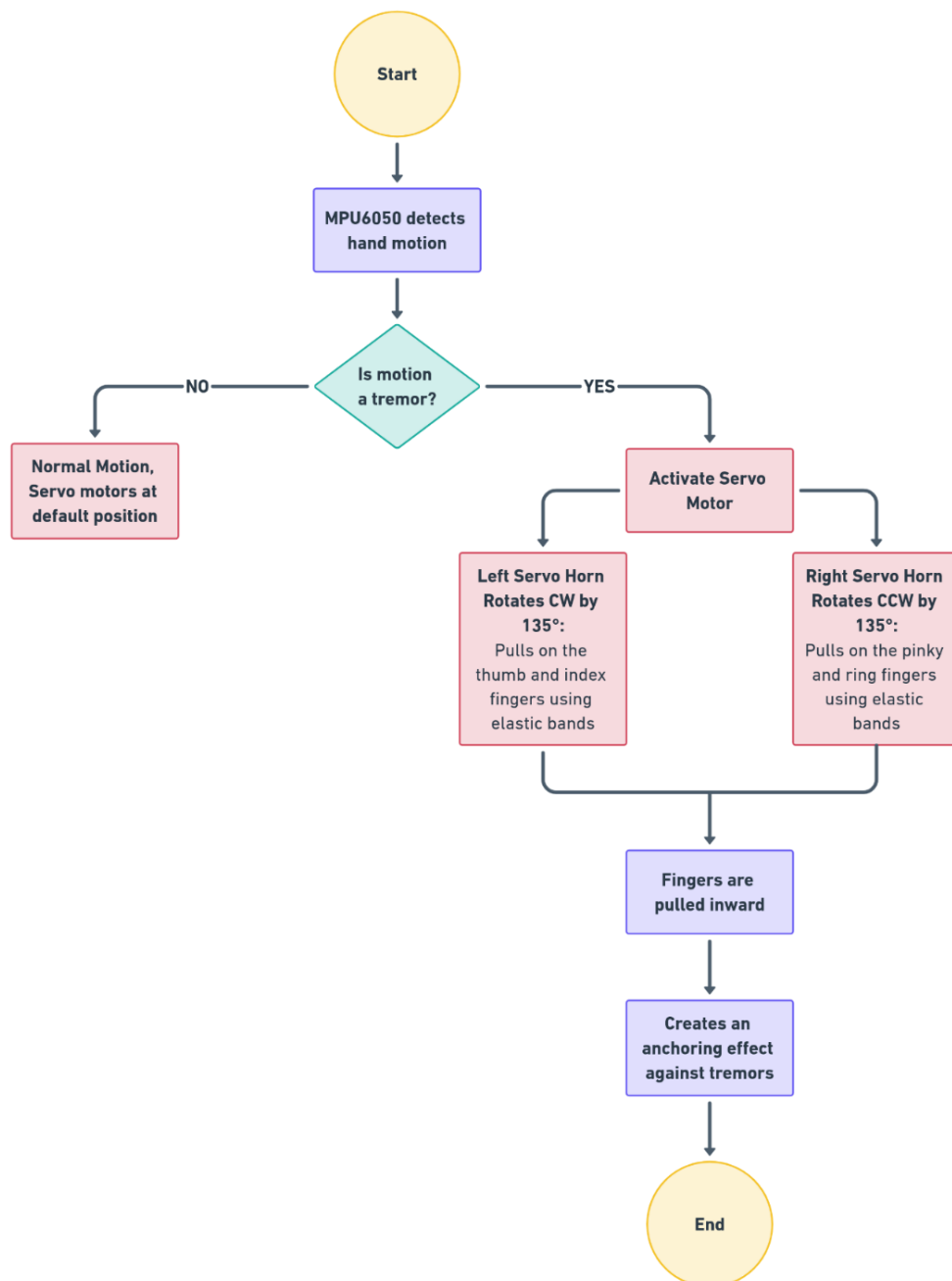


Figure 3: Flowchart illustrating the servo-based tension-anchor damping mechanism

This anchoring effect resists tremor-induced motion, dynamically engaging during tremor events and releasing after stabilization to minimize strain and prevent hand fatigue. A red LED indicator provides immediate visual feedback when tremors are detected.

Power is supplied by a Li-Po battery, with voltage regulated through an LM2596 buck converter to ensure stable system operation.

Software Implementation:

The software for the first prototype was developed using the Arduino IDE, programmed onto the Arduino Nano microcontroller. The main objectives of the software were to continuously detect hand tremors using IMU sensor data and activate the servo-based tension anchor mechanism accordingly. The Arduino code is given below:

```
#include <Wire.h>
#include <Adafruit_MPU6050.h>
#include <Adafruit_Sensor.h>
#include <Servo.h>

// Components
Adafruit_MPU6050 mpu;
Servo servoLeft;
Servo servoRight;

// Pins
const int SERVO_LEFT_PIN = 5;
const int SERVO_RIGHT_PIN = 6;
const int LED_PIN = 7;

// Angles
const int restLeftAngle = 90;
const int restRightAngle = 80;
const int pinchLeftAngle = -20; // Strong CW pull
const int pinchRightAngle = 220; // Strong CCW pull

// Tremor detection
float tremorThreshold = 1.0; // Tune this
unsigned long tremorHoldTime = 1500; // Hold pinch after tremor ends
unsigned long lastTremorTime = 0;

void setup() {
  Serial.begin(115200);
  Wire.begin();
```

```

pinMode(LED_PIN, OUTPUT);
digitalWrite(LED_PIN, LOW);
if (!mpu.begin()) {
    Serial.println("MPU6050 not found!");
    while (1);
}

mpu.setAccelerometerRange(MPU6050_RANGE_8_G);
mpu.setGyroRange(MPU6050_RANGE_500_DEG);
servoLeft.attach(SERVO_LEFT_PIN);
servoRight.attach(SERVO_RIGHT_PIN);
servoLeft.write(restLeftAngle);
servoRight.write(restRightAngle);
delay(1000);
}

void loop() {
    sensors_event_t accel, gyro, temp;
    mpu.getEvent(&accel, &gyro, &temp);

    // Separate tremor axes
    float pitchTremor = abs(gyro.gyro.x); // Up/down
    float rollTremor = abs(gyro.gyro.y); // Left/right

    // Combine both for detection
    float totalTremor = pitchTremor + rollTremor;
    Serial.print("Pitch: "); Serial.print(pitchTremor, 3);
    Serial.print(" | Roll: "); Serial.print(rollTremor, 3);
    Serial.print(" | Total: "); Serial.println(totalTremor, 3);
    if (totalTremor > tremorThreshold) {
        lastTremorTime = millis();
        activatePincher(true, totalTremor);
        Serial.println("Tremor Detected! Pincher active.");
    } else if (millis() - lastTremorTime < tremorHoldTime) {
        activatePincher(true, totalTremor); // Keep holding
        Serial.println("Holding Pincher...");
    } else {
        activatePincher(false, 0);
        Serial.println("Stable. Pincher released.");
    }
    delay(50); // small delay for stability
}

void activatePincher(bool active, float tremorLevel) {
    if (active) {
        // Add oscillation
        float osc = 10 * sin(millis() / 100.0); // +/- 10 degrees
        servoLeft.write(constrain(pinchLeftAngle + osc, 0, 90));
    }
}

```

```
servoRight.write(constrain(pinchRightAngle - osc, 90, 180));  
digitalWrite(LED_PIN, HIGH);  
} else {  
servoLeft.write(restLeftAngle);  
servoRight.write(restRightAngle);  
digitalWrite(LED_PIN, LOW);  
}  
}
```

Chapter 4

Investigation/Experiment, Result, Analysis and Discussion

4.1 Investigation/Experiment

For the EEE499A prototype of the Smart Hand Tremor Suppression and Rehabilitation Glove, the primary goal was to test the glove's functionality in response to simulated tremors. Since actual Parkinsonian tremors were not available for testing at this stage, we simulated hand tremors using a mini vibration motor (MT35). This allowed us to evaluate the system's responsiveness to tremor-like movements and test the core tremor suppression mechanism.

The vibration motor was placed on the glove to generate oscillatory movements mimicking hand tremors. By varying the vibration frequency and intensity, we could simulate different tremor scenarios, ranging from subtle tremors to more pronounced, involuntary movements. This simulation was critical for validating the initial working of the glove and understanding how the tremor detection and suppression system would respond.

The testing was not aimed at real-world validation with Parkinson's patients, but rather at confirming the glove's ability to activate the servo motors and engage the tension-anchor mechanism in response to simulated tremors. The goal was to ensure that the glove could successfully suppress tremor-induced motion in real-time, before moving forward with more advanced testing and refinements.

4.2 Result

The performance of the EEE499A prototype was evaluated based on its ability to respond to simulated tremors, system responsiveness, and energy consumption. Since actual Parkinsonian tremors were not available for testing, we used a mini vibration motor (MT35) to simulate tremor events. Since the vibration capacity of the mini vibration motor was slightly on the weaker side, we had to reduce the detection threshold for this initial test to simulate a tremor

response. This adjustment allowed us to verify the initial working of the glove and ensure it could properly detect and respond to tremor-like movements. The following experimental results summarize the system's performance during these tests.

TABLE II: PERFORMANCE PARAMETERS OF THE FIRST PROTOTYPE DURING SIMULATED TREMOR TESTING

Parameter	Measured value
Tremor Detection Threshold	1.0 deg/s (combined pitch + roll angular velocity)
Average Tremor Detection Response Time	0.48 s
Servo Motor Activation Angle (Left)	135° (clockwise from rest position)
Servo Motor Activation Angle (Right)	135° (counterclockwise from rest position)
Oscillation Range During Activation	$\pm 10^\circ$ (around tensioned position)
Time Holding Tension After Tremor Ends	1.5 s
Regulated Output Voltage (measured using DMM)	5.05 V
Battery Voltage Under Load	7.4 V
Current Draw During Active Suppression (measured using DMM)	~800 mA
Idle Current (no tremor)	~120 mA

In Figure 4, we can see the position change of the servo horns with and without tremor detection. When no tremor is present, the servo motors remain at their resting positions, with minimal tension applied to the elastic bands. However, upon detecting a tremor event, the servo motors rotate 135° (left motor clockwise, right motor counterclockwise), creating the necessary

tension to counteract the tremor and stabilize the hand. The servo horns also demonstrate a slight oscillation of $\pm 10^\circ$ during tremor suppression, which helps in damping minor tremor movements.

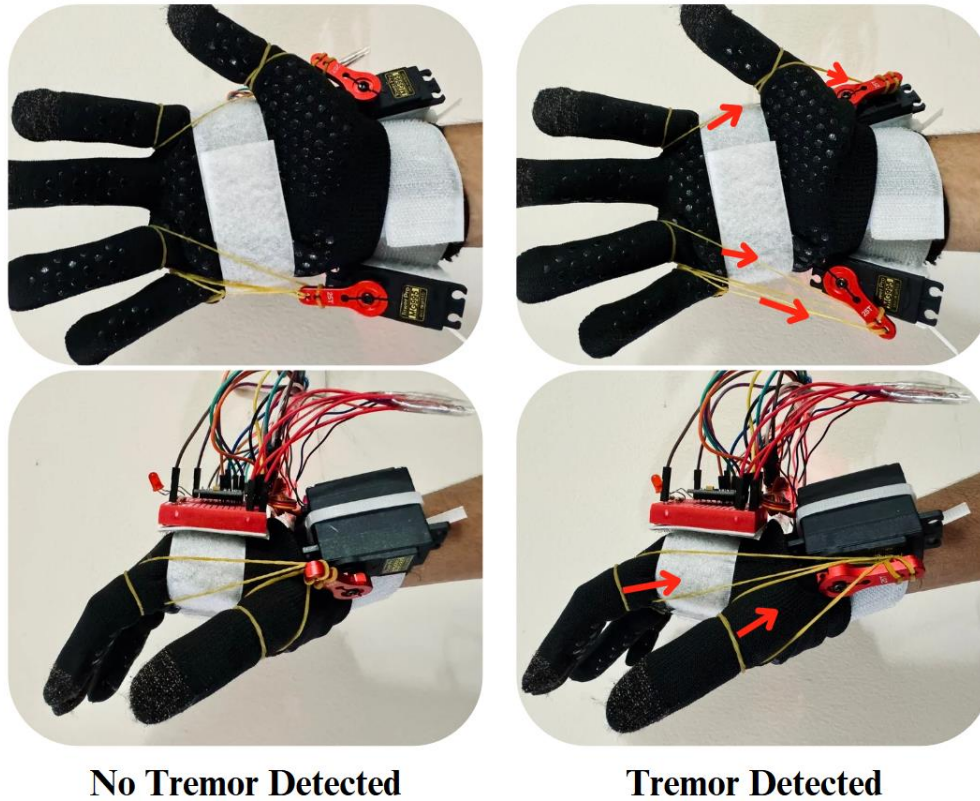


Figure 4: Change in servo horn position with and without tremor detection

4.3 Analysis

The MPU6050 IMU sensor successfully captured the oscillations generated by the vibration motor, demonstrating good sensitivity to tremor-like movements. The tremor detection threshold was set at a combined pitch and roll angular velocity of 1.0 deg/s, which allowed reliable activation during simulation tests. Even with the limited strength of the vibration motor, the system consistently identified tremor-like events, validating the fundamental sensing strategy.

The glove exhibited an average tremor detection response time of 0.48 seconds, which was fast enough to engage suppression before major hand displacement occurred. This relatively low

latency is critical in real tremor suppression applications, where timely intervention can significantly reduce the severity of hand instability.

The mechanical tension-anchor mechanism also performed as intended. Upon tremor detection, the servo motors rotated 135° in opposite directions, pulling on elastic tension bands anchored to the glove. This created a counteracting force across the hand that resisted rapid involuntary movements. The elastic tension effectively damped the simulated tremor oscillations, providing stabilization without rigidly locking the hand. The $\pm 10^\circ$ oscillation range around the tensioned position allowed for some dynamic compliance, enabling better comfort while still maintaining suppression. Additionally, the 1.5-second holding time after tremor cessation helped prevent repetitive activation and deactivation, ensuring smoother mechanical response during intermittent tremors.

The electrical system remained stable throughout the tests. The regulated output voltage stayed consistent at 5.05 V under both idle and active conditions, and the servo motors operated without experiencing voltage drops or system resets. The current draw of approximately 800 mA during active suppression and 120 mA during idle states indicates that the power system was sufficiently sized for the prototype's demands.

Overall, the initial tests demonstrated that both the sensing unit and the mechanical suppression mechanism are functioning as intended, laying a strong foundation for further testing and validation.

4.4 Discussion

The primary goal of this initial testing was to validate that the overall vision of the Smart Hand Tremor Suppression and Rehabilitation Glove could be realized through our proposed design. Using simulated tremors generated by a mini vibration motor, we were able to confirm that both the tremor detection and suppression units worked together effectively. The glove successfully detected tremor-like events and responded with timely mechanical suppression through the tension-anchor mechanism, aligning with the intended operation.

This experiment was not aimed at real-world clinical validation but served as a proof of concept to ensure that the fundamental architecture was sound. It confirmed that the MPU6050 IMU sensor could reliably sense hand motion variations and trigger the servo motors appropriately.

Furthermore, the servo-driven mechanical system proved capable of applying counteracting forces quickly and dynamically enough to potentially stabilize the hand during tremors.

Another important aspect of this testing was verifying the coordination between the detection and suppression units. It was crucial to ensure that sensor data could be processed in real-time and that the mechanical response could follow immediately without lag or errors. The successful integration observed during this phase gives confidence that the system can be further refined and expanded upon in future versions.

While the tremors in this test were artificially generated and relatively simple compared to the complex motion patterns seen in Parkinson's disease, the results are highly promising. Future testing will involve fine-tuning detection thresholds and mechanical responses to better match real patient tremors, but the essential concept—real-time tremor detection followed by mechanical suppression—has been effectively validated through this first prototype.

Chapter 5

Impacts of the Project

5.1 Impact of this project on societal, health, safety, legal and cultural issues

The Smart Hand Tremor Suppression and Rehabilitation Glove project carries significant societal and health implications by offering an accessible, wearable solution for individuals suffering from hand tremors, particularly those affected by neurological disorders such as Parkinson's Disease.

- 1) **Societal Impact:** Hand tremors severely limit the ability to perform daily tasks such as eating, writing, dressing, or using digital devices. By helping stabilize the hand through an intelligent suppression mechanism, this glove can greatly improve users' independence and quality of life. It bridges a gap where traditional treatment methods like medications or surgeries are either too expensive or unavailable, making tremor suppression more equitable and affordable for diverse socioeconomic groups, especially in developing countries like Bangladesh.
- 2) **Health Impact:** The project offers a non-invasive, drug-free alternative to managing tremors, reducing reliance on medications that may have adverse side effects. By physically anchoring the hand during tremor episodes, it may also help prevent accidents (e.g., spills, dropping objects, injuries) associated with uncontrolled hand motion, thereby contributing to overall safety and well-being.
- 3) **Safety Impact:** The system incorporates basic safety considerations in its design by ensuring that servo movement is limited to safe angles, avoiding excessive strain on the hand. Moreover, the glove does not generate any excessive forces, heating, or sharp vibrations that could harm the user.
- 4) **Legal and Ethical Impact:** Since the project focuses on enhancing motor function rather than replacing it, ethical concerns regarding bodily autonomy and dependency are

minimized. Since the glove is an assistive tool and not a medical device, it avoids complicated legal issues. There are also no privacy risks, as the device doesn't collect or share any personal data.

- 5) **Cultural Impact:** In many cultures, physical disabilities are unfortunately stigmatized. By giving individuals a practical way to regain motor control, the project could help reduce social stigma associated with visible tremors, enabling users to feel more confident and socially included.

5.2 Impact of this project on environment and sustainability

The project has been designed with environmental consciousness and sustainability in mind.

Most of the components we used—like the Arduino Nano, MPU6050 sensor, and servo motors—are reusable. If any part fails, it can be replaced without throwing the whole system away, helping to reduce electronic waste. Additionally, because the glove runs on low voltage (5V) and uses small amounts of power, it's very energy-efficient. The materials we are using for tension—elastic bands or springs—are simple, durable, and widely available, reducing the need for frequent replacements.

From a sustainability point of view, this glove offers a more affordable and eco-friendly option compared to expensive surgeries or long-term drug treatments. It provides ongoing benefits without the high financial and environmental costs associated with traditional healthcare. The prototype encourages the practice of repair over replacement—if a servo or elastic wears out, it can easily be swapped instead of discarding the entire glove.

Chapter 6

Project Planning and Budget

6.1 Tentative Project Plan

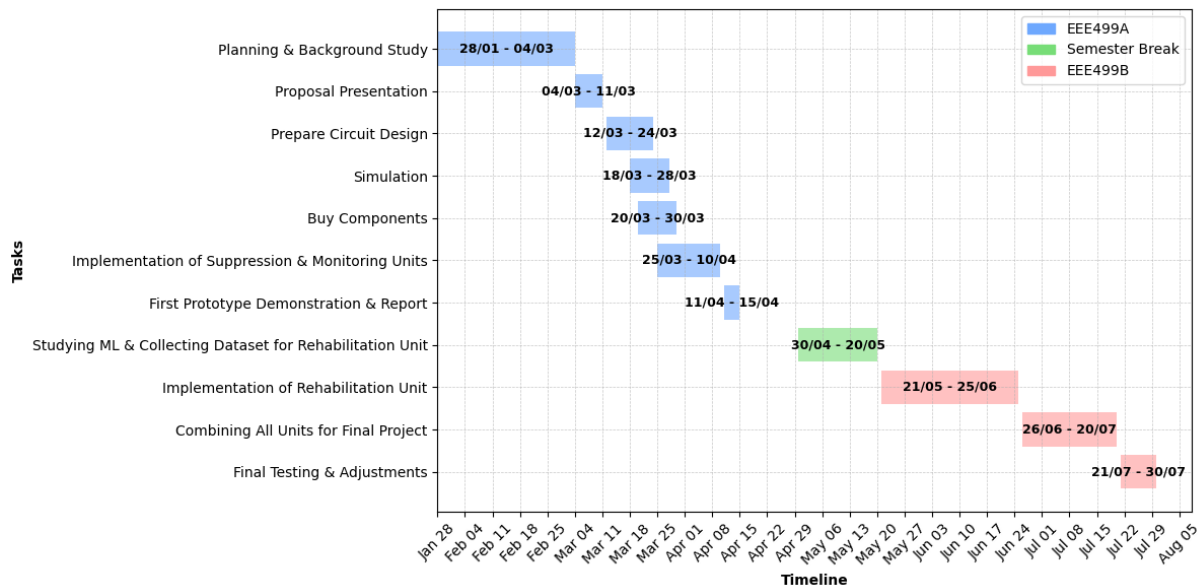


Figure 5: Gantt chart illustrating the proposed project timeline

As shown in Figure 5, the project is divided into three main phases: EEE499A, the Semester Break, and EEE499B, each focusing on different aspects of development. During EEE499A, we successfully designed and built the Tremor Suppression and Monitoring Units. This phase involved developing the servo-based suppression system, integrating motion sensors, and creating Arduino code to enable real-time tremor detection and servo response control. By the end of EEE499A, we achieved our goal of constructing a working prototype that can reliably detect and actively suppress hand tremors. The Semester Break will be utilized for studying machine learning techniques and collecting datasets for the Assisted Rehabilitation Unit. This phase will involve analyzing rehabilitation strategies, preparing data models, and refining the approach for adaptive rehabilitation assistance. In EEE499B, we will focus on implementing the Assisted Rehabilitation Unit and integrating all units into a final, functional system. This includes adding grip-based rehabilitation exercises, real-time feedback mechanisms, and optimizing system performance. Once all units are combined, the final phase will involve

comprehensive testing, adjustments, and validation to ensure that the glove provides both instant tremor suppression and long-term rehabilitation benefits before the final capstone showcase.

6.2 Project Budget

TABLE III: BUDGET TABLE FOR EEE499A PHASE

Phase	Component	Quantity	Price (Tk)
First Purchase	Dual Sensitivity Force Sensor Module – FSR402	1	350
	L7805 Voltage Regulator	2	36
	Servo Motor MG995 180 Degree Rotation	2	738
	MPU6050 IMU Sensor	1	250
	Shipping Cost	-	55
	Total		1429
Second Purchase	25T M3 Metal Servo Arm Horn	2	700
	Shipping Cost	-	60
	Total		760
Third Purchase	Servo Motor MG995 180 Degree Rotation	1	369
	Force Sensitive Resistor 0.5”	1	649
	Arduino Nano	1	385
	Shipping Cost	-	55
	Total		1458
Fourth Purchase	Rocker Switch KCD11	2	12
	Mini Vibration Motor MT35 3V	1	100
	T Connectors Male and Female (Pair)	1	40
	Lipo Battery 1500mAH 7.4 2S	1	1300
	LM2529 DC-DC Buck Converter	1	99
	Shipping Cost	-	55
	Total		1606
Fifth Purchase	Hand Glove	1	340
	Self-adhesive Velcro Tape	1	215
	Shipping Cost	-	170
	Total		715
-	Miscellaneous	-	700

TOTAL SPENT = 6418 Tk

Chapter 7

Conclusions

7.1 Summary

The Smart Hand Tremor Suppression and Rehabilitation Glove project represents a targeted effort to enhance the quality of life for individuals with Parkinson’s disease suffering from debilitating hand tremors. During the EEE499A phase, the focus was placed on the development of a functional, wearable prototype capable of real-time tremor detection and suppression. The glove integrates an MPU6050 inertial measurement unit (IMU) sensor for tremor detection and MG995 servo motors for mechanical suppression through a tension-anchor mechanism. A lightweight algorithm, deployed on an Arduino Nano platform, enabled the system to distinguish tremor-like motion and trigger immediate mechanical suppression with minimal latency.

Initial prototype testing using simulated tremors demonstrated the glove’s ability to reliably detect and counteract involuntary movements, validating the fundamental feasibility of the proposed system. This initial success establishes a strong foundation for the forthcoming EEE499B phase, which will expand the system by incorporating a rehabilitation module to guide users through targeted hand exercises aimed at promoting long-term motor recovery. Additionally, machine learning algorithms will be introduced to enhance tremor classification, personalize suppression strategies, and adapt rehabilitation routines over time. With future integration of IoT-based remote monitoring and adaptive machine learning control, the glove aspires to offer a comprehensive, dual-purpose solution — providing both immediate tremor suppression and sustained rehabilitation.

7.2 Limitations

Despite the successful proof of concept, the current prototype has several limitations. Firstly, the tremor detection and suppression mechanisms were only tested using artificially simulated tremors rather than real Parkinsonian tremors, meaning clinical effectiveness is yet to be validated. The mechanical suppression system, while effective during testing, may need

optimization to adapt to the complex variability in real patient tremors. Additionally, the elastic bands used for the tension-anchor mechanism, while simple and lightweight, may not provide sufficient damping force for stronger or more severe tremor cases. Another limitation is that the glove currently provides suppression only at the prototype level, without full integration of the rehabilitation unit or machine learning-based adaptive features. Furthermore, the device's long-term wearability, battery life, and overall ergonomic comfort were not extensively tested, which are critical factors for real-world usage.

7.3 Future Improvement

Future iterations of the glove should prioritize clinical testing with Parkinson's patients to validate and refine tremor detection thresholds, suppression strategies, and rehabilitation protocols under real-world conditions. The integration of a fully functional rehabilitation unit is essential, enabling guided motor exercises tailored to user-specific needs. Implementation of machine learning algorithms will allow the system to adapt to individual tremor profiles and optimize suppression and rehabilitation in a personalized manner. Mechanically, the elastic bands used in the tension-anchor mechanism will be replaced with springs to provide a stronger, more consistent, and adjustable damping force, better suited to counteract a broader range of tremor severities. Additionally, future designs should aim to improve wearability by making the glove lighter, more flexible, and less obtrusive. Enhancements such as improved battery management, wireless communication modules (e.g., Bluetooth Low Energy or Wi-Fi), and cloud-based monitoring platforms could significantly enhance usability, remote patient monitoring, and long-term data-driven therapy optimization.

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